

What is a Database?

Learning Objectives

- Define a database and explain its purpose in real-world applications
- Differentiate between flat-file and relational databases
- Identify key components of a database management system (DBMS)

Engage (5-7 minutes)

Consider a school with thousands of students—names, grades, attendance, and fees all need to be tracked. Filing cabinets overflow and searching takes forever. A database solves this by storing information electronically so it can be retrieved in seconds.

Explore (10-12 minutes)

Work in groups. One group lists all data a hospital keeps (patients, doctors, appointments), another lists data a bank maintains (accounts, transactions, customers). Compare both lists and identify common patterns: names, IDs, dates, and relationships between records.

Explain (10-12 minutes)

A database is an organized collection of structured data stored electronically and accessible by multiple users. A flat-file database stores all data in a single table, while a relational database splits data across multiple linked tables. A DBMS like MySQL or PostgreSQL is the software that lets users create, read, update, and delete data efficiently.

Elaborate (8-10 minutes)

A local shop currently uses notebooks to track inventory. Explain to the shopkeeper why migrating to a digital database would reduce errors, save time during stock checks, and enable generating sales reports automatically.

Evaluate (5-8 minutes)

Exit ticket: Define “database” in your own words. Name two differences between a flat-file and relational database. Give one example of a DBMS used in industry.